



INFOTECT SOLUTIONS

Snort Intrusion Detection System (IDS) Project Report

Project Title: Implementation and Analysis of Snort IDS

Organization: Infotect Solutions

Prepared by: Nana Kwame Amporful

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Executive Summary

This project presents the implementation and evaluation of a Network Intrusion Detection System (NIDS) using **Snort** for Infotect Solutions. The objective of the project was to install and configure Snort, monitor live network traffic, generate alerts, and analyze suspicious activity through a controlled simulated attack. The project demonstrates how rule-based detection can provide timely and actionable security alerts, helping reduce detection time for common network threats.

The entire exercise was conducted in a controlled lab environment using virtual machines. Snort successfully detected abnormal traffic patterns and generated meaningful alerts, validating its effectiveness as a lightweight and reliable intrusion detection solution.

1. Introduction

Modern organizations rely heavily on networked systems, making continuous traffic monitoring essential for early threat detection. Intrusion Detection Systems (IDS) play a critical role in identifying malicious or abnormal network behavior before it escalates into a full compromise.

This project focuses on deploying Snort as a Network Intrusion Detection System (NIDS) for Infotect Solutions. The work covers installation, configuration, traffic monitoring, alert generation, simulated attack testing, and rule analysis. The goal is to gain practical, hands-

on experience with IDS operations while understanding how Snort rules function in real-world security monitoring.

2. Project Objectives

The main objectives of this project are:

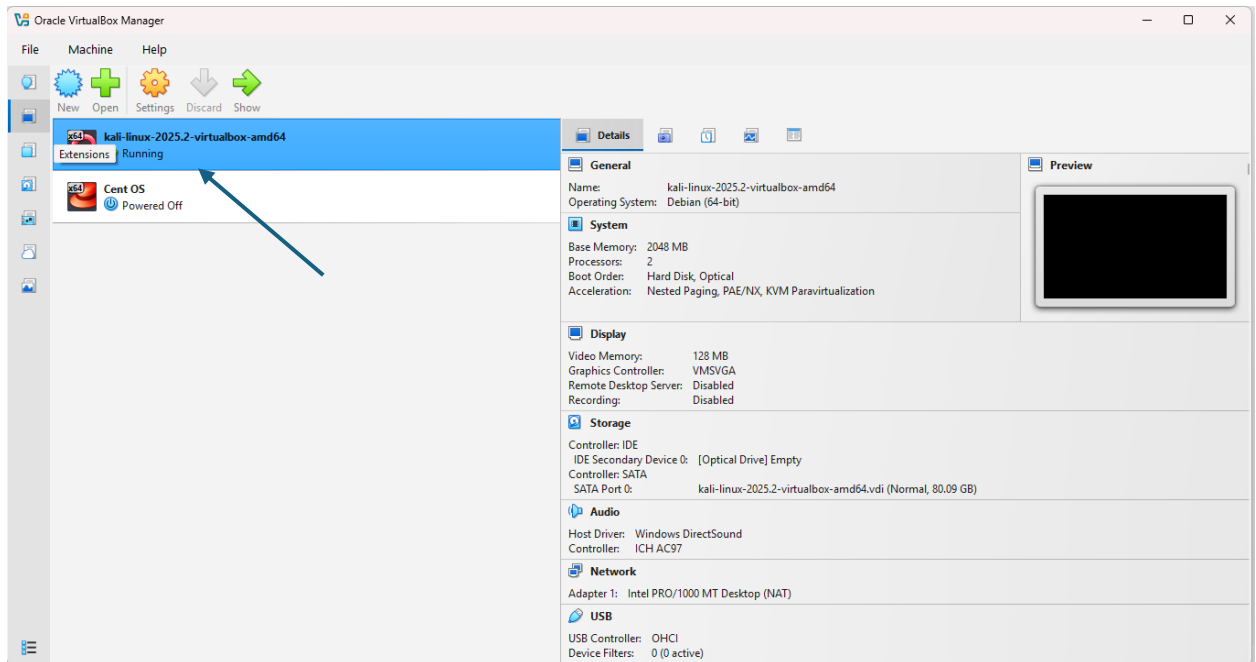
- To install and configure Snort IDS successfully
 - To monitor network traffic in real time
 - To generate and analyze alerts based on detected threats
 - To simulate an attack and observe Snort's response
 - To understand the structure and functionality of Snort rules
-

3. Snort Installation & Setup

3.1 System Requirements

- Operating System: Linux (Ubuntu/Kali Linux)
- Network Interface Card (NIC)
- Internet connection for package installation

Screenshot of the virtual machine(Kali Linux)



3.2 Installation Process

Snort was installed using the system package manager. The installation process included updating system repositories and installing Snort along with its dependencies.

Key steps included:

- Updating system packages

```
(kali@kali)-[~]
$ sudo apt update
Get:1 http://kali.download/kali kali-rolling InRelease [34.0 kB]
Get:2 http://kali.download/kali kali-rolling/main amd64 Packages [20.9 MB]
Get:3 http://kali.download/kali kali-rolling/main amd64 Contents (deb) [52.5 MB]
Get:4 http://kali.download/kali kali-rolling/contrib amd64 Packages [114 kB]
Get:5 http://kali.download/kali kali-rolling/contrib amd64 Contents (deb) [25 5 kB]
Get:6 http://kali.download/kali kali-rolling/non-free amd64 Packages [190 kB]
Get:7 http://kali.download/kali kali-rolling/non-free amd64 Contents (deb) [9 04 kB]
Get:8 http://kali.download/kali kali-rolling/non-free-firmware amd64 Packages [11.8 kB]
Get:9 http://kali.download/kali kali-rolling/non-free-firmware amd64 Contents (deb) [30.0 kB]
Fetched 75.0 MB in 3min 43s (337 kB/s)
1667 packages can be upgraded. Run 'apt list --upgradable' to see them.
```

- Installing Snort

```
kali@kali: ~  
File Actions Edit View Help  
(kali@kali)-[~]  
$ sudo apt install snort  
Upgrading:  
  hwloc          libhwloc15  libxml2-dev  libze1  
  libhwloc-plugins liblzma5    libxml2-utils xz-utils  
  
Installing:  
  snort  
  
Installing dependencies:  
  libdaq3      liblognorm5  oinkmaster  snort-common-libraries  
  libestr0     liblzma-dev  rsyslog     snort-rules-default  
  libfastjson4 libxml2-16   snort-common  
  
Suggested packages:  
  liblzma-doc      rsyslog-elasticsearch  rsyslog-docker  rsyslog-relp  
  rsyslog-doc      rsyslog-kafka          rsyslog-clickhouse  snort-doc  
  rsyslog-mysql    rsyslog-hiredis        rsyslog-openssl  
  rsyslog-pgsql    rsyslog-snmp           | rsyslog-gnutls  
  rsyslog-mongodb  rsyslog-kubernetes    rsyslog-gssapi  
  
Summary:  
  Upgrading: 8, Installing: 12, Removing: 0, Not Upgrading: 1659  
  Download size: 7,906 kB  
  Space needed: 20.3 MB / 63.8 GB available  
  
Continue? [Y/n] y  
Get:1 http://kali.download/kali kali-rolling/main amd64 snort-common-librarie
```

- Verifying Snort installation

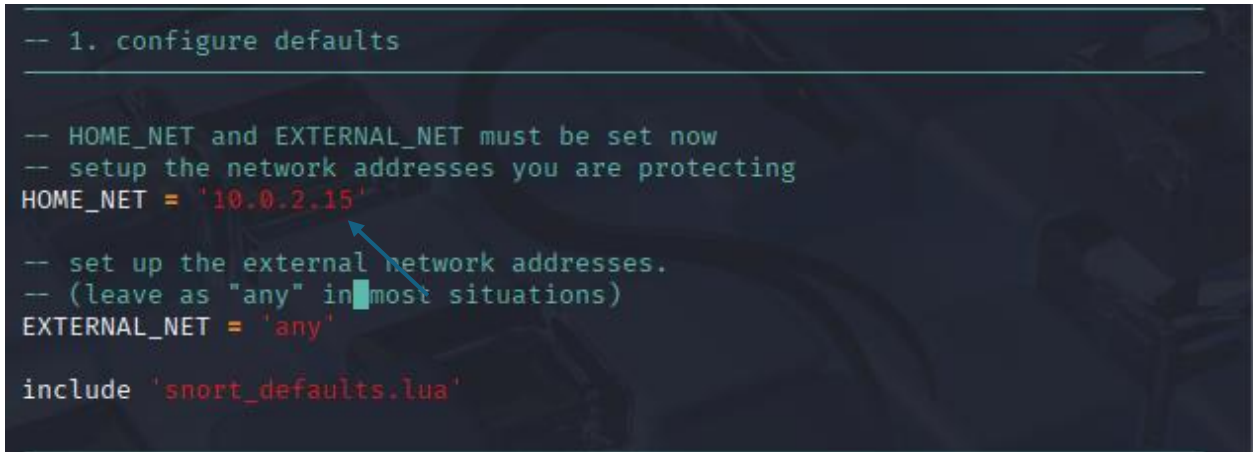
```
(kali@kali)-[~]  
$ snort -v  
o")~  Snort++ 3.10.0.0  
  
Network Policy : policy id 0 :  
Inspection Policy : policy id 0 :  
pcap DAQ configured to passive.  
host_cache  
  memcap: 33554432 bytes  
  
Snort successfully validated the configuration (with 0 warnings).  
o")~  Snort exiting
```

After installation, Snort was configured to run in IDS mode and bind to the active network interface.

3.3 Configuration

The Snort configuration file (snort.conf) was modified to:

- Define the HOME_NET variable



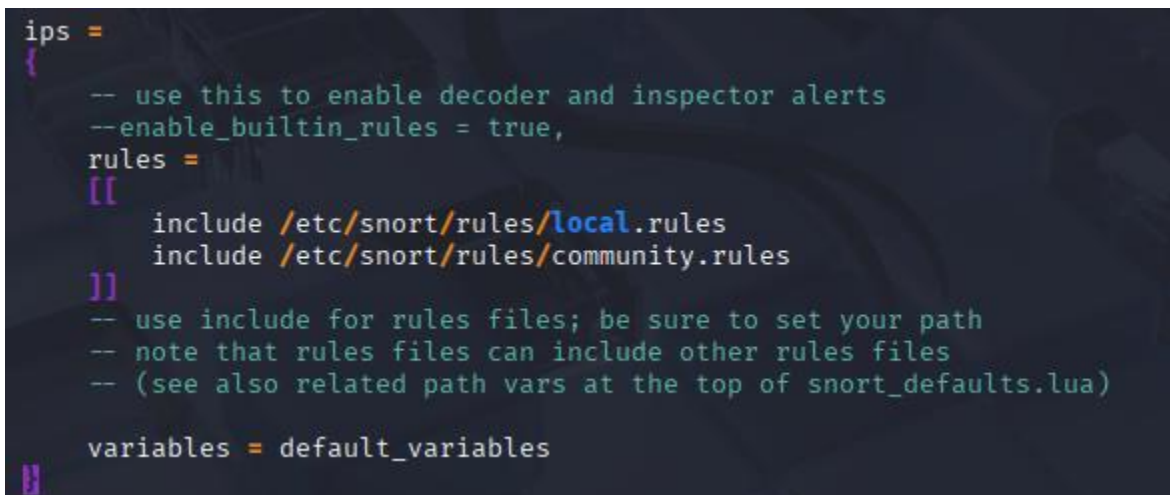
```
-- 1. configure defaults

-- HOME_NET and EXTERNAL_NET must be set now
-- setup the network addresses you are protecting
HOME_NET = '10.0.2.15'

-- set up the external network addresses.
-- (leave as "any" in most situations)
EXTERNAL_NET = 'any'

include 'snort_defaults.lua'
```

- Specify rule paths



```
ips =
{
  -- use this to enable decoder and inspector alerts
  --enable_builtin_rules = true,
  rules =
  [[
    include /etc/snort/rules/local.rules
    include /etc/snort/rules/community.rules
  ]]
  -- use include for rules files; be sure to set your path
  -- note that rules files can include other rules files
  -- (see also related path vars at the top of snort_defaults.lua)

  variables = default_variables
}
```

Proper configuration ensured that Snort could accurately monitor inbound and outbound traffic within the defined network scope.

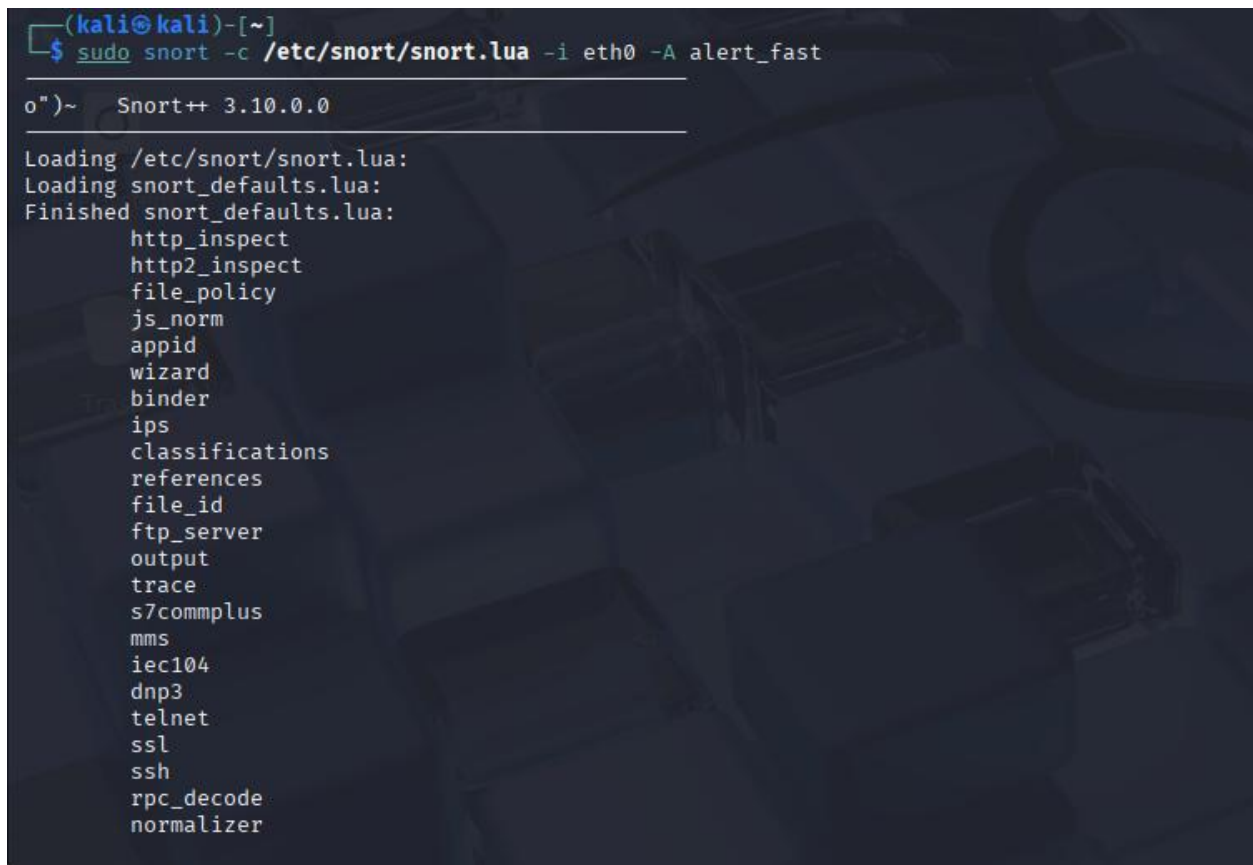
4. Traffic Monitoring and Alert Generation

Snort was executed in network intrusion detection mode to monitor live traffic on the network interface. During this phase:

- Snort captured packets in real time
- Traffic was inspected against predefined rules
- Alerts were generated when suspicious patterns were detected

Snort logs and alert files were reviewed to confirm successful monitoring and alert generation.

📸 Screenshot: Snort running in IDS mode

A screenshot of a terminal window on a Kali Linux system. The prompt is (kali@kali)-[~]. The command executed is \$ sudo snort -c /etc/snort/snort.lua -i eth0 -A alert_fast. The output shows Snort++ 3.10.0.0 and a list of loaded modules: http_inspect, http2_inspect, file_policy, js_norm, appid, wizard, binder, ips, classifications, references, file_id, ftp_server, output, trace, s7complus, mms, iec104, dnp3, telnet, ssl, ssh, rpc_decode, and normalizer.

```
(kali@kali)-[~]  
$ sudo snort -c /etc/snort/snort.lua -i eth0 -A alert_fast  
  
o")~  Snort++ 3.10.0.0  
  
Loading /etc/snort/snort.lua:  
Loading snort_defaults.lua:  
Finished snort_defaults.lua:  
  http_inspect  
  http2_inspect  
  file_policy  
  js_norm  
  appid  
  wizard  
  binder  
  ips  
  classifications  
  references  
  file_id  
  ftp_server  
  output  
  trace  
  s7complus  
  mms  
  iec104  
  dnp3  
  telnet  
  ssl  
  ssh  
  rpc_decode  
  normalizer
```

📸 Screenshot: Generated alerts in Snort alert log


```

pcap DAQ configured to passive.
Commencing packet processing
Retry queue interval is: 200 ms
++ [0] eth0
12/30-19:18:57.393582 [**] [1:1002:1] "ICMP Echo Reply Detected" [**] [Priority: 0] {ICMP} 8.8.8.8 → 10.0.2.15
12/30-19:18:57.393582 [**] [1:1001:1] "ICMP Ping Detected" [**] [Priority: 0] {ICMP} 8.8.8.8 → 10.0.2.15
12/30-19:18:58.392339 [**] [1:1002:1] "ICMP Echo Reply Detected" [**] [Priority: 0] {ICMP} 8.8.8.8 → 10.0.2.15
12/30-19:18:58.392339 [**] [1:1001:1] "ICMP Ping Detected" [**] [Priority: 0] {ICMP} 8.8.8.8 → 10.0.2.15
12/30-19:18:59.393565 [**] [1:1002:1] "ICMP Echo Reply Detected" [**] [Priority: 0] {ICMP} 8.8.8.8 → 10.0.2.15
12/30-19:18:59.393565 [**] [1:1001:1] "ICMP Ping Detected" [**] [Priority: 0] {ICMP} 8.8.8.8 → 10.0.2.15
12/30-19:19:00.403286 [**] [1:1002:1] "ICMP Echo Reply Detected" [**] [Priority: 0] {ICMP} 8.8.8.8 → 10.0.2.15
12/30-19:19:00.403286 [**] [1:1001:1] "ICMP Ping Detected" [**] [Priority: 0] {ICMP} 8.8.8.8 → 10.0.2.15
12/30-19:19:01.400811 [**] [1:1002:1] "ICMP Echo Reply Detected" [**] [Priority: 0] {ICMP} 8.8.8.8 → 10.0.2.15
12/30-19:19:01.400811 [**] [1:1001:1] "ICMP Ping Detected" [**] [Priority: 0] {ICMP} 8.8.8.8 → 10.0.2.15
12/30-19:19:02.403671 [**] [1:1002:1] "ICMP Echo Reply Detected" [**] [Priority: 0] {ICMP} 8.8.8.8 → 10.0.2.15
12/30-19:19:02.403671 [**] [1:1001:1] "ICMP Ping Detected" [**] [Priority: 0] {ICMP} 8.8.8.8 → 10.0.2.15
12/30-19:19:03.402396 [**] [1:1002:1] "ICMP Echo Reply Detected" [**] [Priority: 0] {ICMP} 8.8.8.8 → 10.0.2.15
12/30-19:19:03.402396 [**] [1:1001:1] "ICMP Ping Detected" [**] [Priority: 0] {ICMP} 8.8.8.8 → 10.0.2.15
12/30-19:19:04.429032 [**] [1:1002:1] "ICMP Echo Reply Detected" [**] [Priority: 0] {ICMP} 8.8.8.8 → 10.0.2.15
12/30-19:19:04.429032 [**] [1:1001:1] "ICMP Ping Detected" [**] [Priority: 0] {ICMP} 8.8.8.8 → 10.0.2.15
12/30-19:19:05.440640 [**] [1:1002:1] "ICMP Echo Reply Detected" [**] [Priority: 0] {ICMP} 8.8.8.8 → 10.0.2.15
12/30-19:19:05.440640 [**] [1:1001:1] "ICMP Ping Detected" [**] [Priority: 0] {ICMP} 8.8.8.8 → 10.0.2.15
12/30-19:19:06.434059 [**] [1:1002:1] "ICMP Echo Reply Detected" [**] [Priority: 0] {ICMP} 8.8.8.8 → 10.0.2.15
12/30-19:19:06.434059 [**] [1:1001:1] "ICMP Ping Detected" [**] [Priority: 0] {ICMP} 8.8.8.8 → 10.0.2.15
12/30-19:19:07.439452 [**] [1:1002:1] "ICMP Echo Reply Detected" [**] [Priority: 0] {ICMP} 8.8.8.8 → 10.0.2.15
12/30-19:19:07.439452 [**] [1:1001:1] "ICMP Ping Detected" [**] [Priority: 0] {ICMP} 8.8.8.8 → 10.0.2.15
12/30-19:19:08.447890 [**] [1:1002:1] "ICMP Echo Reply Detected" [**] [Priority: 0] {ICMP} 8.8.8.8 → 10.0.2.15

```

5. Simulated Attack and Alert Analysis

5.1 Simulated Attack

To test Snort's detection capabilities, a simulated attack was carried out. The attack involved generating suspicious network traffic, such as:

- ICMP (Ping) flood

```

pcap DAQ configured to passive.
Commencing packet processing
Retry queue interval is: 200 ms
++ [0] eth0
12/30-19:18:57.393582 [**] [1:1002:1] "ICMP Echo Reply Detected" [**] [Priority: 0] {ICMP} 8.8.8.8 → 10.0.2.15
12/30-19:18:57.393582 [**] [1:1001:1] "ICMP Ping Detected" [**] [Priority: 0] {ICMP} 8.8.8.8 → 10.0.2.15
12/30-19:18:58.392339 [**] [1:1002:1] "ICMP Echo Reply Detected" [**] [Priority: 0] {ICMP} 8.8.8.8 → 10.0.2.15
12/30-19:18:58.392339 [**] [1:1001:1] "ICMP Ping Detected" [**] [Priority: 0] {ICMP} 8.8.8.8 → 10.0.2.15
12/30-19:18:59.393565 [**] [1:1002:1] "ICMP Echo Reply Detected" [**] [Priority: 0] {ICMP} 8.8.8.8 → 10.0.2.15
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12/30-19:19:00.403286 [**] [1:1002:1] "ICMP Echo Reply Detected" [**] [Priority: 0] {ICMP} 8.8.8.8 → 10.0.2.15
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12/30-19:19:01.400811 [**] [1:1002:1] "ICMP Echo Reply Detected" [**] [Priority: 0] {ICMP} 8.8.8.8 → 10.0.2.15
12/30-19:19:01.400811 [**] [1:1001:1] "ICMP Ping Detected" [**] [Priority: 0] {ICMP} 8.8.8.8 → 10.0.2.15
12/30-19:19:02.403671 [**] [1:1002:1] "ICMP Echo Reply Detected" [**] [Priority: 0] {ICMP} 8.8.8.8 → 10.0.2.15
12/30-19:19:02.403671 [**] [1:1001:1] "ICMP Ping Detected" [**] [Priority: 0] {ICMP} 8.8.8.8 → 10.0.2.15
12/30-19:19:03.402396 [**] [1:1002:1] "ICMP Echo Reply Detected" [**] [Priority: 0] {ICMP} 8.8.8.8 → 10.0.2.15
12/30-19:19:03.402396 [**] [1:1001:1] "ICMP Ping Detected" [**] [Priority: 0] {ICMP} 8.8.8.8 → 10.0.2.15
12/30-19:19:04.429032 [**] [1:1002:1] "ICMP Echo Reply Detected" [**] [Priority: 0] {ICMP} 8.8.8.8 → 10.0.2.15
12/30-19:19:04.429032 [**] [1:1001:1] "ICMP Ping Detected" [**] [Priority: 0] {ICMP} 8.8.8.8 → 10.0.2.15
12/30-19:19:05.440640 [**] [1:1002:1] "ICMP Echo Reply Detected" [**] [Priority: 0] {ICMP} 8.8.8.8 → 10.0.2.15
12/30-19:19:05.440640 [**] [1:1001:1] "ICMP Ping Detected" [**] [Priority: 0] {ICMP} 8.8.8.8 → 10.0.2.15
12/30-19:19:06.434059 [**] [1:1002:1] "ICMP Echo Reply Detected" [**] [Priority: 0] {ICMP} 8.8.8.8 → 10.0.2.15
12/30-19:19:06.434059 [**] [1:1001:1] "ICMP Ping Detected" [**] [Priority: 0] {ICMP} 8.8.8.8 → 10.0.2.15
12/30-19:19:07.439452 [**] [1:1002:1] "ICMP Echo Reply Detected" [**] [Priority: 0] {ICMP} 8.8.8.8 → 10.0.2.15
12/30-19:19:07.439452 [**] [1:1001:1] "ICMP Ping Detected" [**] [Priority: 0] {ICMP} 8.8.8.8 → 10.0.2.15
12/30-19:19:08.447890 [**] [1:1002:1] "ICMP Echo Reply Detected" [**] [Priority: 0] {ICMP} 8.8.8.8 → 10.0.2.15

```

- Port scanning using Nmap


```

12/30-19:21:06.433393 [**] [1:1018:1] "DNS Response Detected" [**] [Priority: 0] {UDP} 192.168.0.1:53 → 10.0.2.15:41281
12/30-19:21:06.439392 [**] [1:1018:1] "DNS Response Detected" [**] [Priority: 0] {UDP} 192.168.0.1:53 → 10.0.2.15:41281
12/30-19:21:06.537574 [**] [1:1018:1] "DNS Response Detected" [**] [Priority: 0] {UDP} 192.168.0.1:53 → 10.0.2.15:59333
12/30-19:21:06.555284 [**] [1:1020:1] "DNS over TCP Detected" [**] [Priority: 0] {TCP} 10.0.2.15:49869 → 216.58.215.142:53
12/30-19:21:06.598861 [**] [1:1002:1] "ICMP Echo Reply Detected" [**] [Priority: 0] {ICMP} 216.58.215.142 → 10.0.2.15
12/30-19:21:06.598861 [**] [1:1001:1] "ICMP Ping Detected" [**] [Priority: 0] {ICMP} 216.58.215.142 → 10.0.2.15
12/30-19:21:08.363687 [**] [1:1020:1] "DNS over TCP Detected" [**] [Priority: 0] {TCP} 10.0.2.15:49871 → 216.58.215.142:53
12/30-19:21:10.945515 [**] [1:1020:1] "DNS over TCP Detected" [**] [Priority: 0] {TCP} 10.0.2.15:49873 → 216.58.215.142:53

```

5.2 Detection and Alert Generation

Snort successfully detected the simulated attack traffic and generated alerts indicating:

- Source IP address
- Destination IP address
- Type of attack detected
- Timestamp of the event

5.3 Alert Analysis

The generated alerts were analyzed to understand:

- The nature of the attack
- Which rule was triggered
- The severity level of the alert

This analysis demonstrated Snort's effectiveness in identifying and reporting malicious network activities.

 *Screenshot: Snort alert triggered by simulated attack*

```

12/30-19:19:18.464187 [**] [1:1002:1] "ICMP Echo Reply Detected" [**] [Priority: 0] {ICMP} 8.8.8.8 → 10.0.2.15
12/30-19:19:18.464187 [**] [1:1001:1] "ICMP Ping Detected" [**] [Priority: 0] {ICMP} 8.8.8.8 → 10.0.2.15
12/30-19:19:19.467955 [**] [1:1002:1] "ICMP Echo Reply Detected" [**] [Priority: 0] {ICMP} 8.8.8.8 → 10.0.2.15
12/30-19:19:19.467955 [**] [1:1001:1] "ICMP Ping Detected" [**] [Priority: 0] {ICMP} 8.8.8.8 → 10.0.2.15
12/30-19:19:20.471864 [**] [1:1002:1] "ICMP Echo Reply Detected" [**] [Priority: 0] {ICMP} 8.8.8.8 → 10.0.2.15
12/30-19:19:20.471864 [**] [1:1001:1] "ICMP Ping Detected" [**] [Priority: 0] {ICMP} 8.8.8.8 → 10.0.2.15
12/30-19:19:21.474037 [**] [1:1002:1] "ICMP Echo Reply Detected" [**] [Priority: 0] {ICMP} 8.8.8.8 → 10.0.2.15
12/30-19:19:21.474037 [**] [1:1001:1] "ICMP Ping Detected" [**] [Priority: 0] {ICMP} 8.8.8.8 → 10.0.2.15
12/30-19:19:22.479142 [**] [1:1002:1] "ICMP Echo Reply Detected" [**] [Priority: 0] {ICMP} 8.8.8.8 → 10.0.2.15
12/30-19:19:22.479142 [**] [1:1001:1] "ICMP Ping Detected" [**] [Priority: 0] {ICMP} 8.8.8.8 → 10.0.2.15
12/30-19:19:23.479573 [**] [1:1002:1] "ICMP Echo Reply Detected" [**] [Priority: 0] {ICMP} 8.8.8.8 → 10.0.2.15
12/30-19:19:23.479573 [**] [1:1001:1] "ICMP Ping Detected" [**] [Priority: 0] {ICMP} 8.8.8.8 → 10.0.2.15
12/30-19:19:25.515100 [**] [1:1002:1] "ICMP Echo Reply Detected" [**] [Priority: 0] {ICMP} 8.8.8.8 → 10.0.2.15
12/30-19:19:25.515100 [**] [1:1001:1] "ICMP Ping Detected" [**] [Priority: 0] {ICMP} 8.8.8.8 → 10.0.2.15
12/30-19:19:26.518674 [**] [1:1002:1] "ICMP Echo Reply Detected" [**] [Priority: 0] {ICMP} 8.8.8.8 → 10.0.2.15
12/30-19:19:26.518674 [**] [1:1001:1] "ICMP Ping Detected" [**] [Priority: 0] {ICMP} 8.8.8.8 → 10.0.2.15
12/30-19:19:27.520589 [**] [1:1002:1] "ICMP Echo Reply Detected" [**] [Priority: 0] {ICMP} 8.8.8.8 → 10.0.2.15
12/30-19:19:27.520589 [**] [1:1001:1] "ICMP Ping Detected" [**] [Priority: 0] {ICMP} 8.8.8.8 → 10.0.2.15
12/30-19:19:28.515895 [**] [1:1002:1] "ICMP Echo Reply Detected" [**] [Priority: 0] {ICMP} 8.8.8.8 → 10.0.2.15
12/30-19:19:28.515895 [**] [1:1001:1] "ICMP Ping Detected" [**] [Priority: 0] {ICMP} 8.8.8.8 → 10.0.2.15
12/30-19:19:29.535173 [**] [1:1002:1] "ICMP Echo Reply Detected" [**] [Priority: 0] {ICMP} 8.8.8.8 → 10.0.2.15
12/30-19:19:29.535173 [**] [1:1001:1] "ICMP Ping Detected" [**] [Priority: 0] {ICMP} 8.8.8.8 → 10.0.2.15
12/30-19:19:30.535450 [**] [1:1002:1] "ICMP Echo Reply Detected" [**] [Priority: 0] {ICMP} 8.8.8.8 → 10.0.2.15
12/30-19:19:30.535450 [**] [1:1001:1] "ICMP Ping Detected" [**] [Priority: 0] {ICMP} 8.8.8.8 → 10.0.2.15
12/30-19:21:06.433393 [**] [1:1018:1] "DNS Response Detected" [**] [Priority: 0] {UDP} 192.168.0.1:53 → 10.0.2.15:41281
12/30-19:21:06.439392 [**] [1:1018:1] "DNS Response Detected" [**] [Priority: 0] {UDP} 192.168.0.1:53 → 10.0.2.15:41281
12/30-19:21:06.537574 [**] [1:1018:1] "DNS Response Detected" [**] [Priority: 0] {UDP} 192.168.0.1:53 → 10.0.2.15:59333
12/30-19:21:06.555284 [**] [1:1020:1] "DNS over TCP Detected" [**] [Priority: 0] {TCP} 10.0.2.15:49869 → 216.58.215.142:53
12/30-19:21:06.598861 [**] [1:1002:1] "ICMP Echo Reply Detected" [**] [Priority: 0] {ICMP} 216.58.215.142 → 10.0.2.15
12/30-19:21:06.598861 [**] [1:1001:1] "ICMP Ping Detected" [**] [Priority: 0] {ICMP} 216.58.215.142 → 10.0.2.15
12/30-19:21:08.363687 [**] [1:1020:1] "DNS over TCP Detected" [**] [Priority: 0] {TCP} 10.0.2.15:49871 → 216.58.215.142:53
12/30-19:21:10.945515 [**] [1:1020:1] "DNS over TCP Detected" [**] [Priority: 0] {TCP} 10.0.2.15:49873 → 216.58.215.142:53

```

6. Understanding of Snort Rules

Snort rules form the core of its detection engine. Each rule consists of two main parts:

6.1 Rule Header

The rule header defines:

- Action (alert, log, pass)
- Protocol (TCP, UDP, ICMP)
- Source and destination IP addresses
- Source and destination ports

6.2 Rule Options

The rule options provide detailed information such as:

- Message to display when rule is triggered
- Content to match in packets
- Rule ID (SID)
- Revision number

Example Snort Rule:

```
alert icmp any any -> $HOME_NET any (msg:"ICMP Ping Detected"; sid:1000001; rev:1;)
```

This rule generates an alert whenever an ICMP ping is detected targeting the home network.

 *Screenshot: Custom Snort rule configuration*

```
# $Id: local.rules,v 1.11 2004/07/23 20:15:44 bmc Exp $
#
# LOCAL RULES
#
# This file intentionally does not come with signatures.  Put your local
# additions here.

# ICMP Rules
alert icmp any any -> $HOME_NET any ( msg:"ICMP Ping Detected"; sid:1001; rev:1; )
alert icmp any any -> $HOME_NET any ( msg:"ICMP Echo Reply Detected"; itype:0; sid:1002; rev:1; )
alert icmp any any -> $HOME_NET any ( msg:"ICMP Destination Unreachable"; itype:3; sid:1003; rev:1; )
alert icmp any any -> $HOME_NET any ( msg:"ICMP Redirect Detected"; itype:5; sid:1004; rev:1; )
alert icmp any any -> $HOME_NET any ( msg:"ICMP Time Exceeded"; itype:11; sid:1005; rev:1; )

# HTTP/Web Traffic Rules
alert tcp any any -> $HOME_NET 80 ( msg:"HTTP Request Detected"; sid:1006; rev:1; )
alert tcp any any -> $HOME_NET 443 ( msg:"HTTPS Connection Attempt"; sid:1007; rev:1; )
alert tcp any any -> $HOME_NET 80 ( msg:"Possible SQL Injection Attempt"; content:"union select"; sid:1008; rev:1; )
alert tcp any any -> $HOME_NET 80 ( msg:"Possible XSS Attempt"; content:"<script>"; sid:1009; rev:1; )
alert tcp any any -> $HOME_NET 80 ( msg:"Directory Traversal Attempt"; content:"../"; sid:1010; rev:1; )
alert tcp any any -> $HOME_NET 80 ( msg:"Possible Command Injection"; content:"|3B|"; sid:1011; rev:1; )
alert tcp any any -> $HOME_NET 80 ( msg:"HTTP POST Request"; content:"POST"; sid:1012; rev:1; )
alert tcp any any -> $HOME_NET 8080 ( msg:"HTTP Alternate Port Access"; sid:1013; rev:1; )

# SSH Rules
alert tcp any any -> $HOME_NET 22 ( msg:"SSH Connection Attempt"; sid:1014; rev:1; )
alert tcp any any -> $HOME_NET 22 ( msg:"Possible SSH Brute Force"; detection_filter:track by_src, count 5, seconds 60; sid:1015; rev:1; )

G Help      O Write Out  F Where Is   K Cut        T Execute    C Location   M-U Undo     M-A Set Mark M-T To Bracket M-B Previous
X Exit      R Read File  N Replace    u Paste      o Justify    G Go To Line M-E Redo     M-C Copy     B Where Was  F Next
```

7. Conclusion

This project successfully demonstrated the installation, configuration, and use of Snort IDS for network security monitoring at Infotect Solutions. Through real-time traffic monitoring, alert generation, and simulated attacks, Snort proved to be an effective tool for detecting suspicious network activities.

The project also enhanced understanding of Snort rules and their role in intrusion detection. Snort remains a valuable and cost-effective solution for organizations seeking to strengthen their network security posture.

8. References

- [Snort - Open-Source Network Intrusion Detection & Prevention System || Cisco Talos Intelligence Group - Comprehensive Threat Intelligence](#)
 - [Snort 3 Rule Writing Guide - Snort 3 Rule Writing Guide](#)
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End of Report